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CAT 1

Using examples, describe software testing life cycle

Software testing life cycle is the step-by-step procedure of inspecting software with defined activities to achieve specific goals before publishing. It is verifying and validating a program.

1. Software Requirements Specification – identifying and analyzing what to be tested.
2. Planning – defining scope, objectives, strategy and resources to be used as well as estimating related costs.
3. Developing test case - series of required steps that need to be followed to test the software. Each step may contain specific instructions and their expected results.
4. Environment setup – hardware/software configuration.
5. Execution – implementing tests and collecting logs as well as fixing bugs and retesting.
6. Closing – documentation of findings and conclusions.

For instance, a software product (Fitness App) has been submitted for testing.

1. Software Requirements Specification

- Review documented requirements such as workout logging, calorie tracking, and GPS route mapping.
- Constraints: Android/iOS compatibility, response time under 2 seconds.

2. Planning

- Scope: Collecting workout data (logging exercises, calories, GPS routes).
- Strategy: Automate regression tests for workout frequency changes; keep manual tests for GPS accuracy.
- Deliverable: Test plan with timelines, tools (Appium, JUnit), and resource planning.

- Tools: radiometer, weighing scale

3. Developing Test Cases

- TC01: Verify logging a 30-minute run saves correctly in workout history.
- TC02: Verify calorie calculation matches expected formula.
- TC03: Verify GPS route is mapped accurately on the app screen.
- Deliverable: Test case repository with expected results.

4. Environment Setup

- Install app on Android emulator and iOS simulator.
- Connect to backend API with sample user accounts.
- Load test data (workouts, calorie tables).
- Deliverable: Validated test environment ready for execution.

5. Execution

- TC01 passes; TC02 fails (calorie mismatch defect logged); TC03 fails (GPS skips points defect logged).
- Deliverable: Test execution report with pass/fail status and defect logs.

6. Closing

- Summarize results, analyze metrics, document lessons.
- Example: 95% test cases passed, 5 defects logged and fixed.
- Deliverable: Test summary report and sign-off for release.

Reference

1. Jomo Kenyatta University of Agriculture and Technology. (2017). *ICS 2302 Software Engineering I: Lesson 3 – Phases in the Development of Software*. School of Open, Distance and eLearning (SODEL).
2. Sommerville, I. (2016). *Software Engineering* (10th ed.). Pearson Education.